

Transformation Patterns from BFO-based to UFO-based concepts

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1 Introduction

This document clarifies the Transformation Patterns to support the reuse of concepts defined as instances or as subclasses of Basic Formal Ontology (BFO) in ontologies based on Unified Foundational Ontology (UFO).

Transformation Patterns are a practical tool to guide modelers that use the OWL language¹ in their ontology engineering practices. For the purposes of this work, we utilize the OWL versions of BFO² and UFO³ that correspond to the version formally recognized by the ISO/IEC 21838-2 standard, and that are accessible in GitHub⁴.

For each Transformation Pattern, we present the definition for BFO source and UFO target hierarchy of classes stated in the ontologies' available documentation on GitHub. Additional examples (when available) and comments are provided to clarify the Transformation Pattern.

2 Entity and Continuant

The first Transformation Patterns aim the **BFO:Entity** and **BFO:Continuant** classes. Figure 1 illustrates, on the left side, the excerpt of the BFO ontology covered by the Transformation Patterns defined in this document, along with the excerpt of OWL implementation of UFO ontology (on the right side) for which such Transformation Patterns will result in target hierarchies.

¹<https://www.w3.org/OWL/>

²<https://github.com/BFO-ontology/BFO-2020/tree/master/21838-2>

³<https://github.com/nemo-ufes/gufo>

⁴Because **gUFO** is the lightweight OWL implementation of UFO, we use the **gUFO** notation to refer to UFO's classes as formalized in OWL, and **UFO** to refer to the foundational ontology.

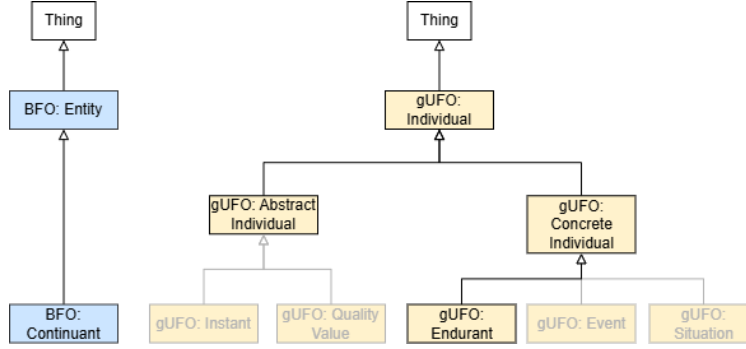


Figure 1: Excerpts of the taxonomies of Basic Formal Ontology (BFO) and Unified Foundational Ontology (UFO) covered by the Transformations Patterns #1 and #2

2.1 BFO:Entity

2.1.1 Definitions

BFO:Entity: (Elucidation) An entity is anything that exists or has existed or will exist.

gUFO:Individual: An entity that (unlike a **gufo:Type**) cannot be instantiated.

gUFO:AbstractIndividual: A **gUFO:Individual** that does not exist in space-time in the same way as a **gufo:ConcreteIndividual** does. A **gufo:AbstractIndividual** has no spatiotemporal qualities in its own right. Hence, it does not make sense to ask how much space it now occupies (...) and when it was created or destroyed.

gUFO:ConcreteIndividual: A **gufo:Individual** that exists in space-time.

2.1.2 Examples and Considerations

According to gUFO’s documentation, **gUFO:Type** is *[a]n entity that may be instantiated by (or predicated over) other entities. Types encompass what we often call ‘sorts’, ‘kinds’, ‘categories’, etc. Examples include the kind ‘Person’, the event type ‘Earthquake’, and the abstract individual type ‘NaturalNumber’. Relations, such as ‘marriedTo’ and ‘olderThan’, are also considered as types.*

Instances of **gufo:Type** are classes, and should specialize the taxonomy of individuals of gUFO. For example, “Person” is a **gufo:Type** (more specifically a **gufo:Kind**), specializing **gufo:Object**. The mechanism that allows for this is called punning in OWL 2.

Since BFO does not support a theory of types, the Transformation Patterns are aimed at UFO’s taxonomy of Individuals.

In UFO, Individual is *[a]n entity that (unlike a gufo:Type) cannot be instantiated. Individuals may be either concrete (e.g., the Earth, Mick Jagger, Brazil, the 1985 Mexico City*

Earthquake) or abstract (e.g., the number two, the proposition that ‘three is a prime number’). In the philosophical literature, this notion is usually referred to as a *particular*. UFO distinguishes its Individuals between Concrete and Abstract Individuals, a distinction that is not supported by BFO.

A Concrete Individual is *an Individual that exists in space-time*. This definition is closer to the one adopted for Entity in BFO (“anything that exists”), with the restriction that UFO is accounting for “entities that cannot be instantiated” existing in space-time, while BFO accounts for “anything”, as long as it exists. This may implicate in modelling choices when accounting for concepts that do not fulfill UFO’s “concreteness”.

These considerations motivate the first Transformation Pattern, that includes the assessment of possible intended abstract and concrete characteristics in domain concepts.

2.1.3 Transformation Pattern

Transformation Pattern #1 BFO:Entity has target Hierarchy gUFO:Individual, and may be further specialized according to the transformation patterns:

- *Does the Entity exist in space-time either as an object that endures time (for example, a car, a person, a marriage, a thought) or as an event that unfolds in time (such as the sunrise)?*

BFO:Entity has target Hierarchy gUFO:ConcreteIndividual

- *Does the Entity lack spatiotemporal qualities, or is the Entity not existing in space-time, in the sense that it cannot be created or destroyed (such as the number ten, or a proposition)?*

BFO:Entity has target Hierarchy gUFO:AbstractIndividual

2.2 BFO:Continuant

2.2.1 Definitions

BFO:Continuant: (Elucidation) A continuant is an entity that persists, endures, or continues to exist through time while maintaining its identity.

gUFO:Endurant: *a Concrete Individual that endures in time and may change qualitatively while keeping its identity.*

2.2.2 Examples and Considerations

From BFO’s documentation, examples of Continuants are [a] *human being; a tennis ball; a cave; a region of space; someone’s temperature.*

UFO’s examples mention *ordinary objects of everyday experience* (a person, a house, a car); *reified relationships* (a marriage, a rental contract, and a person’s love for another) and *existentially-dependent aspects of objects* (a car’s weight, a person’s language skills, and a house’s color).

UFO’s documentation states *endurants* are termed *continuants* in the philosophical literature, supporting the second transformation pattern.

2.2.3 Transformation Pattern

Transformation Pattern #2 BFO:Continuant has target Hierarchy gUFO:Endurant.

3 Independent Continuant Hierarchy

The next set of Transformation Patterns are those that contemplate concepts defined as instances or as subclasses of the BFO:IndependentContinuant class and its subclasses, namely: BFO:ImmaterialEntity, BFO:ContinuantFlatBoundary, BFO:Site, BFO:SpatialRegion, BFO:MaterialEntity, BFO:Object, BFO:ObjectAggregate and BFO:FiatObjectBoundary. Figure 2 illustrates, on the left side, the excerpt of the BFO ontology covered by the Transformation Patterns defined in this document, along with the excerpt of OWL’s implementation of UFO ontology (on the right side) for which such Transformation Patterns will result in target hierarchies.

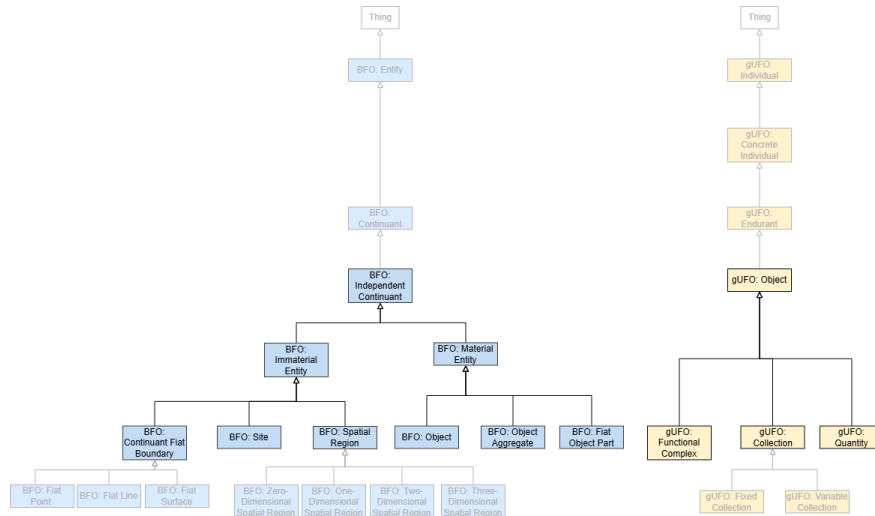


Figure 2: Excerpts of the taxonomies of Basic Formal Ontology (BFO) and Unified Foundational Ontology (UFO) covered by the Transformations Patterns #3, #4, #5, #6, #7, #8, #9, #10 and #11

For each Transformation Pattern, we present the definition for BFO source and UFO target hierarchy of classes stated in the ontologies’ available documentation on GitHub.

Additional examples from the Sustainability Core Ontology (SCO)⁵ and GeoCore Ontology⁶ (when available) and comments are provided to clarify the Transformation Pattern.

3.1 BFO:IndependentContinuant

3.1.1 Definitions

BFO:IndependentContinuant: b is an independent continuant =Def b is a continuant & there is no c such that b specifically depends on c or b generically depends on c

gUFO:Object: A **gufo:Endurant** that does not depend on another endurant for its existence (excluding its essential parts and aspects).

3.1.2 Examples and Considerations

BFO: Independent Continuant: An atom; a molecule; an organism; a heart; a chair; the bottom right portion of a human torso; a leg; the interior of your mouth; a spatial region; an orchestra

UFO: Object: ordinary physical entities, such as a dog, a house, a tomato, a car, Alan Turing, but also socially-defined entities such as The Rolling Stones, the European Union, the Brazilian 1988 Constitution.

In UFO, an Endurant can be distinguished according its dependency on another Endurant. The Endurant hierarchy is divided in Object, or *[a]n endurant that that does not depend on another endurant for its existence (excluding its essential parts and aspects)* or Aspect, the *endurant that depends on at least one other concrete individual for its existence*.

This notion of non-dependence is closely related to BFO's definition of Independent Continuants, hence the Transformation Pattern below.

3.1.3 Transformation Pattern

Transformation Pattern #3 BFO:IndependentContinuant *has target Hierarchy* gUFO:Object.

3.2 BFO:ImmaterialEntity

3.2.1 Definitions

BFO:ImmaterialEntity: b is an immaterial entity =Def b is an independent continuant which is such that there is no time t when it has a material entity as continuant part

⁵<https://github.com/gioUbbiali/Sustainability-Core-Ontology>

⁶<https://github.com/BDI-UFRGS/GeoCoreOntology/tree/master>

gUFO:Object: A **gufo:Endurant** that does not depend on another endurant for its existence (excluding its essential parts and aspects).

3.2.2 Examples and Considerations

Immaterial entities in BFO were created to express surfaces, lines and points that describe boundary and adjacency relationships of material physical entities and spaces. UFO, however, does not specify a theory for relationships of localization. This has implications on how immaterial entities are transformed: because they are usually mapped to UFO's Objects, the Transformation Patterns include some restrictions to observe the different behaviors of these entities.

The constitution of UFO's Objects include the notion of "Substantial", which generalizes both objects and amounts of matter. However, the examples mention *physical entities* and do not specifically describe independent endurants that have no material parts - hence the lack of examples from UFO documentation in the Transformation Patterns regarding Immaterial Entities.

Immaterial Entities modeled in BFO have UFO's Objects as target hierarchy, provided that we observe the specific transformations of **BFO:ContinuantFiatBoundary**, **BFO:Site** and **BFO:SpatialRegion**. The proposed Transformation Patterns do not consider examples that fall outside the specificity of either of these three classes.

3.2.3 Transformation Pattern

Transformation Pattern #4 **BFO:ImmaterialEntity** has target Hierarchy **gUFO:Object**, considering the following constraints:

- If the source ontology has concepts modeled under the **BFO:MaterialEntity** hierarchy and concepts under the **BFO:ImmaterialEntity** hierarchy, they must be mapped to *disjoint* subclasses of **gUFO:Object**;
- **BFO:ContinuantFiatBoundary**, **BFO:Site** and **BFO:SpatialRegion** are transformed to distinct *disjoint* subclasses of **gUFO:Object**.

3.3 BFO:ContinuantFiatBoundary

3.3.1 Definitions

BFO:ContinuantFiatBoundary: (Elucidation) A continuant fiat boundary *b* is an immaterial entity that is of zero, one or two dimensions & such that there is no time *t* when *b* has a spatial region as continuant part & whose location is determined in relation to some material entity

gUFO:Object: A **gufo:Endurant** that does not depend on another endurant for its existence (excluding its essential parts and aspects).

3.3.2 Examples and Considerations

BFO: Continuant Fiat Boundary: (Arp et al. 2015) the surface of the Earth, the boundary of a portion of airspace into which one aircraft is allowed to fly.

GeoCore: geological boundary (2D): a fiat surface that is *located in* the external surface of a geological object.

SCO: system boundary: A fiat surface of a system.

BFO's theory for granularity admits that object boundaries (such as the surface of a tomato) exist *because of the existence of the objects they bound*, as seen when these objects are viewed from a perspective of medium-sized object granularity. The Transformation Pattern for Continuant Fiat Boundaries considers this restriction, admitting parthood relations between a `gUFO:Object` derived from a Material Entity and the respective `gUFO:Objects` that constitute its boundaries.

Regarding UFO's notion of Quantities, we assume it is very unlikely that one would have the intention to model the (moving) boundary of a portion of matter. This would demand a deeper philosophical approach, with considerations regarding object-boundary relationships' domain and range, which should be restricted. An obvious question that arises from this is "which implications for a model would arise from considering that the boundary of the portion of wine (a `gUFO:Object`) is `ObjectProperPart` of the portion of wine (a `gUFO:Quantity`)"? That discussion motivates the restriction on the following Transformation Pattern.

3.3.3 Transformation Pattern

Transformation Pattern #5 `BFO:ContinuantFiatBoundary` has target *Hierarchy* `gUFO:Object`, considering the following constraints:

- `BFO:ContinuantFiatBoundary`, `BFO:Site` and `BFO:SpatialRegion` are transformed to distinct *disjoint* subclasses of `gUFO:Object`;
- `BFO:ContinuantFiatBoundary` has target *hierarchy* a (`gUFO:Object` and not a `gUFO:Quantity`) that *isObjectProperPartOf* some other (`gUFO:Object` and not a `gUFO:Quantity`) from which it is disjoint.

3.4 BFO:Site

3.4.1 Definitions

BFO:Site: (Elucidation) A site is a three-dimensional immaterial entity whose boundaries either (partially or wholly) coincide with the boundaries of one or more material entities or have locations determined in relation to some material entity

gUF0:Object: A **gufo:Endurant** that does not depend on another endurant for its existence (excluding its essential parts and aspects).

3.4.2 Examples and Considerations

BFO: Site: (Arp et al. 2015) nasal cavity, the Suez Canal, the interior of your office, your eye socket

As for Continuant Fiat Boundaries, examples help clarify that Sites are also defined in relation to a material entity. According to (Arp et al., 2015) [a] *site exists because there is some material entity in relation to which it is defined, providing, for example, the floor and walls and ceiling that allow containment by forming the retainer of the site. (...) But the site itself, while it exists because of this retainer, does not contain the retainer as part. The site is, rather, the hole that is contained by the retainer (...)*. This motivates an analogous Transformation Pattern.

3.4.3 Transformation Pattern

Transformation Pattern #6 **BFO:Site** has target *Hierarchy* **gUF0:Object**, considering the following constraints:

- **BFO:ContinuantFiatBoundary**, **BFO:Site** and **BFO:SpatialRegion** are transformed to distinct *disjoint* subclasses of **gUF0:Object**;
- **BFO:Site** has target *hierarchy* a **gUF0:Object** that can be related to some other **gUF0:Object** representing the **BFO:MaterialEntity** that constitutes its retainer, via an *isObjectProperPartOf* relation.

3.5 BFO:SpatialRegion

3.5.1 Definitions

BFO:SpatialRegion: (Elucidation) A spatial region is a continuant entity that is a continuant part of the spatial projection of a portion of spacetime at a given time

gUF0:Object: A **gufo:Endurant** that does not depend on another endurant for its existence (excluding its essential parts and aspects).

3.5.2 Examples and Considerations

SCO: system spatial region: A three-dimensional spatial region of a system.

Spatial regions are a part of space, occupied by both material and immaterial entities. Sites may coincide with some spatial region, but spatial regions are not dependent upon a

retainer. For example, a ship’s cargo hold is a moving site, one that will cycle through a continuous multiplicity of spatial regions.

While independent continuants may have qualities (such as shape, size, mass and density, reflectance, electric charge and so forth), spatial regions, belong to a peculiar (“abstract”) sort: no realizable inhere in them. Despite having qualities of shape and size, but *the primary BFO relation here is one of instantiation between a spatial region instance and the corresponding spatial region universal* (Arp et al., 2015).

In accepting Sites into its ontology in addition to Spatial Regions, BFO acknowledges two distinct location relations involving independent continuants: containment and location. Those relations are not natively defined in UFO, which can motivate the creation of additional relations that are not included in the Transformation Pattern below.

3.5.3 Transformation Pattern

Transformation Pattern #7 BFO:SpatialRegion has target Hierarchy gUFO:Object, considering the following constraint:

- BFO:ContinuantFiatBoundary, BFO:Site and BFO:SpatialRegion are transformed to distinct *disjoint* subclasses of gUFO:Object.

3.6 BFO:MaterialEntity

3.6.1 Definitions

BFO:MaterialEntity: (Elucidation) A material entity is an independent continuant that at all times at which it exists has some portion of matter as continuant part

gUFO:Object: A gufo:Endurant that does not depend on another endurant for its existence (excluding its essential parts and aspects).

gUFO:Quantity: A complex gufo:Object that is a maximally-connected portion of stuff. A gufo:Quantity has a fixed constitution, and thus, removing or adding a sub-quantity would result in a different quantity.

gUFO:FunctionalComplex: A complex gufo:Object whose parts (components) play different roles in its composition.

3.6.2 Examples and Considerations

BFO: Material Entity: A human being; the undetached arm of a human being; an aggregate of human beings

UFO: Object: *Ordinary physical entities* (a house, a tomato, a car, Alan Turing), *socially-defined entities* (The Rolling Stones, the European Union, the Brazilian 1988 Constitution).

UFO: Quantity: the portion of wine in a wine tank, a lump of clay, the gold that constitutes a wedding ring.

UFO: Functional Complex: a person with the various organs (heart, brain, lungs, etc.) playing different roles; a scrum team composed by people playing the roles of scrum master, product owner, developer, etc.

GeoCore: earth material: a natural amount of matter *generated by* some geological process.

SCO: system: A material entity consisting of multiple components that are causally integrated.

SCO: system component: A material entity that belongs to a system.

SCO: system environment: A material entity that surrounds and is distinct and disjoint from a system.

SCO: system environment component: A material entity that belongs to a system environment.

As discussed earlier, UFO does not explicitly defines independent endurants that do not have material parts. That "collapses" the Transformation Patterns both for Material Entities and Objects (in the BFO sense), as shown below.

On the other hand, BFO does not allow for the co-existence of distinct entities in the same spatial and temporal locations, and one of the reasons is that it does not include a theory that supports portions of matter. In this sense, BFO and UFO have very different perspectives on how a sum of bricks can or cannot differ from a wall. In the case of material entities, the Transformation Patterns motivate the assessment of their constitution and complexity, resulting in different specializations of UFO's Objects.

Depending on the intended meaning of a given Material Entity (or Object), it can be transformed either to a Quantity, an Object or a FunctionalComplex. The choice of modelling a (BFO-derived) Object as a Quantity is based on the object's constitution: UFO supports the idea of (self-connected) amounts of matter that are maximal relative to the sort of causal unity manifested. In the case of Functional Complexes, the choice relies on the intention of the model to account for Object's components that play different roles in their composition. For example, we may want to model the spaceship's engine, fuel tank and other parts. A person could be considered a Functional Complex with their various organs (heart, brain, lungs, etc.) playing different roles.

3.6.3 Transformation Pattern

Transformation Pattern #8 `BF0:MaterialEntity` has target hierarchy `gUFO:Object`, and may be further specialized according to transformation patterns:

- *Is the Material Entity a maximally self-connected portion of stuff, such as the portion of wine in a wine tank?*

`BF0:MaterialEntity` has target hierarchy `gUFO:Quantity`

- *Are the Material Entity's components modeled as playing different roles in the Material Entity's composition, such as the various organs of a person (person being the object), or the engine of a car (the object being the car)?*

`BF0:MaterialEntity` has target hierarchy `gUFO:FunctionalComplex`

- *Is the Material Entity a component that plays a role in a different complex Material Entity, such as the brain of a person?*

`BF0:MaterialEntity` has target hierarchy `gUFO:Object` that can be related to a complex `gUFO:FunctionalComplex` that represents the `BF0:MaterialEntity` it is a component of via an *isComponentOf* relation.

3.7 BF0:Object

3.7.1 Definitions

BF0:Object: (Elucidation) An object is a material entity which manifests causal unity & is of a type instances of which are maximal relative to the sort of causal unity manifested

gUFO:Object: A `gufo:Endurant` that does not depend on another endurant for its existence (excluding its essential parts and aspects).

gUFO:Quantity: A complex `gufo:Object` that is a maximally-connected portion of stuff. A `gufo:Quantity` has a fixed constitution, and thus, removing or adding a sub-quantity would result in a different quantity.

gUFO:FunctionalComplex: A complex `gufo:Object` whose parts (components) play different roles in its composition.

3.7.2 Examples and Considerations

BFO: Object: (Arp et al., 2015) a single cell, a spaceship, an egg, a planet

UFO: Object: *Ordinary physical entities* (such as a dog, a house, a tomato, a car, Alan Turing), *socially-defined entities* (The Rolling Stones, the European Union, the Brazilian 1988 Constitution).

UFO: Quantity the portion of wine in a wine tank, a lump of clay, the gold that constitutes a wedding ring.

UFO: Functional Complex: a person with the various organs (heart, brain, lungs, etc.) playing different roles; a scrum team composed by people playing the roles of scrum master, product owner, developer, etc.

GeoCore: geological object: An object that is *generated by* some geological process and has at least one part *constituted by* some earth material that is not an earth fluid.

As discussed earlier, UFO does not explicitly defines independent endurants that do not have material parts. That "collapses" the Transformation Patterns both for Material Entities and Objects (in the BFO sense), as shown below. The same considerations made for Material Entities remain valid for the following Transformation Pattern.

3.7.3 Transformation Pattern

Transformation Pattern #9 BFO:Object has target hierarchy gUFO:Object, and may be further specialized according to transformation patterns:

- *Is the Object a maximally self-connected portion of stuff, such as the portion of wine in a wine tank?*

BFO:Object has target hierarchy gUFO:Quantity

- *Are the the Object's components modeled as playing different roles in the Object's composition, such as the various organs of a person (person being the object), or the engine of a car (the object being the car)?*

BFO:Object has target hierarchy gUFO:FunctionalComplex

- *Is the Object a component that plays a role in a different complex Material Entity, such as the brain of a person?*

BFO:Object) has target hierarchy gUFO:Object that can be related to a complex gUFO:FunctionalComplex that represents the BFO:MaterialEntity or BFO:Object it is a component of via an *isComponentOf* relation.

3.8 BFO:ObjectAggregate

3.8.1 Definitions

BFO:ObjectAggregate: (Elucidation) An object aggregate is a material entity consisting exactly of a plurality (≥ 1) of objects as member parts which together form a unit

gUFO:Object: A **gufo:Endurant** that does not depend on another endurant for its existence (excluding its essential parts and aspects).

gUFO:FunctionalComplex: A complex **gufo:Object** whose parts (components) play different roles in its composition.

gUFO:Collection: A complex **gufo:Object** whose parts (the members of the collection) have a uniform structure (i.e., members are conceived as playing the same role in the collection). Collections may have a fixed or variable membership, which can be asserted using **gufo:FixedCollection** and **gufo:VariableCollection** respectively.

3.8.2 Examples and Considerations

BFO: Object Aggregate: (Arp et al., 2015) A heap of stones, a group of commuters on the subway, a flock of geese, the collection of patients in a hospital

UFO: Collection a deck of cards, a pile of bricks, a forest (conceived as a collection of trees), and a group of people.

UFO: Functional Complex a person with the various organs (heart, brain, lungs, etc.) playing different roles; a scrum team composed by people playing the roles of scrum master, product owner, developer, etc.

Object Aggregates consist of a collection (or *plurality*) of objects as member parts which together form a unit, and whose parts are exactly exhausted by the objects that form this collection. This notion is the equivalent of the UFO concept of Collections, a *complex gufo:Object whose parts (the members of the collection) have a uniform structure (i.e., members are conceived as playing the same role in the collection)*.

3.8.3 Transformation Pattern

Transformation Pattern #10 **BFO:ObjectAggregate** has target hierarchy **BFO:Object**, specialized according to the following transformation patterns:

- *Does the Object Aggregate have a uniform structure (i.e., members are conceived as playing the same role in the collection)?*

BFO:ObjectAggregate has target hierarchy **gUFO:Collection**

- *Are the Object Aggregate's components modelled as playing different roles in its composition, such as the various organs of a person (person being the object), or the engine of a car (the object being the car)?*

BFO:ObjectAggregate has target hierarchy **gUFO:FunctionalComplex**

3.9 BFO:FiatObjectPart

3.9.1 Definitions

BFO:FiatObjectPart: (Elucidation) A fiat object part *b* is a material entity & such that if *b* exists then it is continuant part of some object *c* & demarcated from the remainder of *c* by one or more fiat surfaces

gUFO:Object: A **gufo:Endurant** that does not depend on another endurant for its existence (excluding its essential parts and aspects).

gUFO:Quantity: A complex **gufo:Object** that is a maximally-connected portion of stuff. A **gufo:Quantity** has a fixed constitution, and thus, removing or adding a sub-quantity would result in a different quantity.

3.9.2 Examples and Considerations

BFO: Fiat Object Part: (Arp et al., 2015) your upper torso, the handle of a solid metal spoon, the western hemisphere of Earth

UFO: Object: *Ordinary physical entities* (such as a dog, a house, a tomato, a car, Alan Turing), *socially-defined entities* (The Rolling Stones, the European Union, the Brazilian 1988 Constitution).

UFO: Quantity the portion of wine in a wine tank, a lump of clay, the gold that constitutes a wedding ring.

In BFO there is the concept of Fiat Object Part, which could constrain the existence of a Functional Complex as Objects composed of Fiat Object Parts. However, there is a problem with the spatial conditioning of Fiat Object Parts, which are demarcated from the remainder of the object by fiat surfaces. Under that spatial conditioning, this view would not account for UFO cases such as a scrum team – a team member would not be demarcated from the rest of the team by a surface. Hence the following Transformation Pattern, in which the main consideration is constitution of the Object of which the entity is a Fiat Object Part (allowing for parts of objects and of amounts of matter).

3.9.3 Transformation Pattern

Transformation Pattern #11 **BFO:FiatObjectPart** has target hierarchy **BFO:Object**, specialized according to the following transformation patterns:

- *Is the Fiat Object Part a portion of a bigger maximally self-connected portion of stuff, such as the quantity of water in a wine glass (as a sub-quantity of the wine in that glass)?*

`BFO:FiatObjectPart` has target hierarchy `gUFO:Quantity` that can be related to another `gUFO:Quantity` representing the `BFO:Object` it is a part of, via a *isSubQuantityOf* relation.

4 Specifically and Generically Dependent Continuant hierarchy

The next set of Transformation Patterns supports the transformation of concepts defined as instances or as subclasses of the `BFO:SpecificallyDependentContinuant` and `BFO:GenericallyDependentContinuant` classes and its subclasses, namely: `BFO:Quality`, `BFO:RelationalQuality`, `BFO:RealizableEntity`, `BFO:Role`, `BFO:Disposition` and `BFO:Function`. Figure 3 illustrates, on the left side, the excerpt of the BFO ontology covered by the Transformation Patterns defined in this document, along with the excerpt of the OWL implementation of UFO ontology (on the right side) for which such Transformation Patterns will result in target hierarchies.

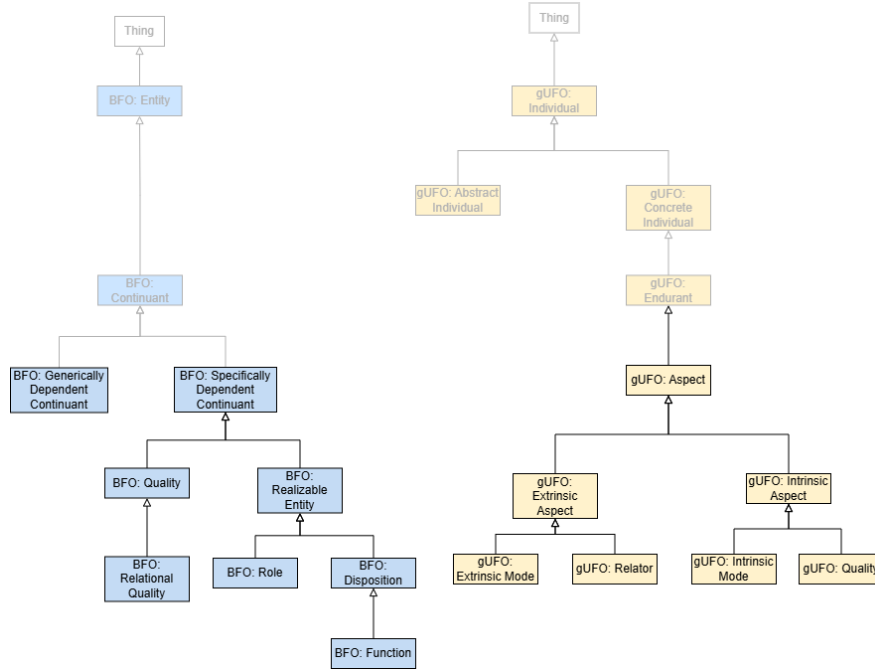


Figure 3: Excerpts of the taxonomies of Basic Formal Ontology (BFO) and Unified Foundational Ontology (UFO) covered by the Transformations Patterns #12, #13, #14, #15, #16, #17, #18, #19

For each Transformation Pattern, we present the definition for BFO source and UFO target hierarchy of classes stated in the ontologies' available documentation on GitHub. Additional examples from the Sustainability Core Ontology (SCO) and GeoCore Ontology (when available) and comments are provided to clarify the Transformation Pattern.

4.1 BFO:SpecificallyDependentContinuant

4.1.1 Definitions

BFO:SpecificallyDependentContinuant: b is a specifically dependent continuant =Def b is a continuant & there is some independent continuant c which is not a spatial region & which is such that b specifically depends on c

gUFO:Aspect: A **gufo:Endurant** that depends on at least one other concrete individual for its existence. A **gufo:Aspect** is a characteristic or trait of a concrete individual that is itself conceived as an individual.

gUFO:ExtrinsicAspect: A **gufo:Aspect** that depends on one or more concrete individuals.

gUFO:IntrinsicAspect: A **gufo:Aspect** that depends on a single concrete individual in which it inheres.

4.1.2 Examples and Considerations

BFO: Specifically Dependent Continuant: (with multiple bearers) John's love for Mary; the ownership relation between John and this statue; the relation of authority between John and his subordinates

UFO: Aspect: *intrinsic physical aspects* (such as the Moon's mass, Lassie's fur color); *mental dispositions* (such as Bob's math skills, his belief that the number one is odd); *relational aspects* (John's love for Mary and the marriage between John and Mary).

SCO: complex system property: (Comment) This class is constructed as a grouping for complex system properties, such as emergence and self-organization, in order to facilitate class retrieval and visualization.

A **UFO:Aspect** is an *Endurant* that depends on at least one other concrete individual for its existence. It is a characteristic or trait of a concrete individual that is itself conceived as an individual.

This correlates with BFO's definition of Specifically Dependent Continuant. However, BFO and UFO propose very different views for dependent entities. In BFO they are classified according to their *realization* or not on dependent continuants through processes. In UFO, the classification is based on how many individuals that aspect depends on (simultaneously) to exist. This impacts how the following Transformation Patterns target different (disjoint) hierarchies.

4.1.3 Transformation Patterns

Transformation Pattern #12 `BF0:SpecificallyDependentContinuant` has target *Hierarchy* `gUFO:Aspect`, considering the following constraints and further specializations:

- *Does the Specifically Dependent Continuant depend on the single individual in which it inheres (such as the propensity of glass to break)?*

`BF0:SpecificallyDependentContinuant` has target hierarchy `gUFO:IntrinsicAspect`

- *Does the Specifically Dependent Continuant exist because of other individual's existence, even if that individual is not represented in the model (for example, the role of a molecule to be used as medicine, or the socially-defined role of a student)?*

`BF0:SpecificallyDependentContinuant` has target hierarchy `gUFO:ExtrinsicAspect`

4.2 BF0:Quality

4.2.1 Definitions

BF0:Quality: (Elucidation) A quality is a specifically dependent continuant that, in contrast to roles and dispositions, does not require any further process in order to be realized

gUFO:Aspect: A `gufo:Endurant` that depends on at least one other concrete individual for its existence.

gUFO:Quality: A `gufo:IntrinsicAspect` that is measurable by some value spaces. A quality may be used to compare individuals, on the basis of the value it takes in a certain quality space (for instance, a position within the RGB spectrum).

4.2.2 Examples and Considerations

BFO: Quality: (Arp et al. 2015) the mass of this kidney, the color of this portion of blood, RGB 990033 dark red

UFO: Aspect: *intrinsic physical aspects* (such as the Moon's mass, Lassie's fur color); *mental dispositions* (such as Bob's math skills, his belief that the number one is odd); *relational aspects* (John's love for Mary and the marriage between John and Mary).

UFO: Quality: the weight of a person, the name of organization, the color of a car, and the duration of a concert.

Both definitions of Quality are very similar, which is reinforced by the examples provided by the two foundational ontologies. However, UFO classifies its Aspects according to how many individuals it depends on (simultaneously) to exist. This justifies the restriction of

the Transformation Pattern below, accounting for Qualities that may inhere in multiple individuals simultaneously.

4.2.3 Transformation Patterns

Transformation Pattern #13 BFO:Quality *has target Hierarchy* gUFO:Aspect, and can be further specialized according to the following transformation patterns:

- *Does this Quality concern single individuals, in which it inheres?*
BFO:Quality *has target hierarchy* gUFO:Quality

4.3 BFO:RelationalQuality

4.3.1 Definitions

BFO:RelationalQuality: b is a relational quality =Def b is a quality & there exists c and d such that c and d are not identical & b specifically depends on c & b specifically depends on d

gUFO:ExtrinsicAspect: A gufo:Aspect that depends on one or more concrete individuals.

gUFO:ExtrinsicMode: A gufo:ExtrinsicAspect that inheres in a concrete individual and depends on others for its existence.

gUFO:Relator: A gufo:ExtrinsicAspect that connects (involves, mediates) two or more concrete individuals. Relators are reified relationships composed of reciprocal extrinsic modes.

4.3.2 Examples and Considerations

BFO: Relational Quality: (Arp et al. 2015) a marriage bond, an instance of love, being a parent

UFO: Extrinsic Aspect: *reified relationships* (John and Mary’s marriage, Mary’s employment contract at Nasa); *parts of those relationships* (John’s obligations towards Mary in the scope of the marriage); *one-sided relationships* (John’s admiration for Obama).

GeoCore: Geological Contact: a relational quality that *inheres in* two distinct geological object that are *externally connected with* (i.e., their external boundaries are physically adjacent)

4.3.3 Transformation Patterns

Transformation Pattern #14 BFO:RelationalQuality *has target Hierarchy* gUFO:ExtrinsicAspect, considering the following transformation patterns:

- *Does this Relational Quality involve and/or mediates two or more concrete individuals (as reified relationships composed of reciprocal extrinsic modes)?*

BFO:RelationalQuality *has target hierarchy* gUFO:Relator

- *Does this Relational Quality inheres in a concrete individual and depends on others for its existence (as reified one-sided relationships)?*

BFO:RelationalQuality *has target hierarchy* gUFO:ExtrinsicMode

4.4 BFO:RealizableEntity

4.4.1 Definitions

BFO:RealizableEntity: (Elucidation) A realizable entity is a specifically dependent continuant that inheres in some independent continuant which is not a spatial region & which is of a type some instances of which are realized in processes of a correlated type

gUFO:Aspect: A gufo:Endurant that depends on at least one other concrete individual for its existence. A gufo:Aspect is a characteristic or trait of a concrete individual that is itself conceived as an individual.

gUFO:ExtrinsicAspect: A gufo:Aspect that depends on one or more concrete individuals.

gUFO:IntrinsicAspect: A gufo:Aspect that depends on a single concrete individual in which it inheres.

4.4.2 Examples and Considerations

BFO: Realizable Entity: The role of being a doctor; the role of this boundary to delineate where Utah and Colorado meet; the function of your reproductive organs; the disposition of your blood to coagulate; the disposition of this piece of metal to conduct electricity

As discussed earlier, UFO does not take classify its Aspects (or individuals that depend on at least one other concrete individual for its existence) according to their realization through processes. As a consequence, the Transformation Pattern may result in loss of specificity.

4.4.3 Transformation Patterns

Transformation Pattern #15 *BF0:RealizableEntity has target hierarchy gUF0:Aspect, considering the following constraints and possible specializations:*

- If the source ontology has concepts modeled as **BF0:Quality** (or its subclasses) and as **BF0:RealizableEntity** (or its subclasses) that are transformed to **gUF0:Aspect**, they must be *disjoint* subclasses of **gUF0:Aspect**;
- *Does the Realizable Entity depend on the single individual in which it inheres (such as the propensity of glass to break)?*

BF0:RealizableEntity has target hierarchy gUF0:IntrinsicAspect

- *Does the Realizable Entity exist because of other individual's existence, even if that individual is not represented in the model (for example, the role of a molecule to be used as medicine or the socially-defined role of a student)?*

BF0:RealizableEntity (BF0:Disposition) has target hierarchy gUF0:ExtrinsicAspect

4.5 BF0:Disposition

4.5.1 Definitions

BF0:Disposition: (Elucidation) A disposition *b* is a realizable entity such that if *b* ceases to exist then its bearer is physically changed & *b*'s realization occurs when and because this bearer is in some special physical circumstances & this realization occurs in virtue of the bearer's physical make-up

gUF0:Aspect: A **gufo:Endurant** that depends on at least one other concrete individual for its existence. A **gufo:Aspect** is a characteristic or trait of a concrete individual that is itself conceived as an individual.

gUF0:ExtrinsicAspect: A **gufo:Aspect** that depends on one or more concrete individuals.

gUF0:IntrinsicAspect: A **gufo:Aspect** that depends on a single concrete individual in which it inheres.

4.5.2 Examples and Considerations

BFO: Disposition: (Arp et al. 2015) a hemophiliac's disposition to bleed abnormally, the disposition of a magnet to attract iron

SCO: perspective: A disposition whose realization leads the complex system component or the complex system environment component to (re)position itself within and regarding the complex system or complex system environment it belongs to (or a part of it) and possibly act accordingly.

While Transforming a concept that is defined as **BF0:Disposition**, it is important to know that UFO does not classify its dependent individuals according to their realization through processes. As a consequence, the Transformation Pattern may result in loss of specificity, analogously to the Transformation Pattern for Realizable Entities.

4.5.3 Transformation Patterns

Transformation Pattern #16 **BF0:Disposition** *has target hierarchy* **gUFO:Aspect**, considering the following constraints and possible specializations:

- If the source ontology has concepts modeled as **BF0:Quality** (or its subclasses) and as **BF0:RealizableEntity** (or its subclasses) that are transformed to **gUFO:Aspect**, they must be *disjoint* subclasses of **gUFO:Aspect**;
- *Does the Disposition depend on the single individual in which it inheres (such as the propensity of glass to break)?*

BF0:Disposition *has target hierarchy* **gUFO:IntrinsicAspect**

- *Does the Disposition exist because of other individual's existence, even if that individual is not represented in the model (for example, the role of a molecule to be used as medicine or the socially-defined role of a student)?*

BF0:Disposition *has target hierarchy* **gUFO:ExtrinsicAspect**

4.6 BF0:Role

4.6.1 Definitions

BF0:Role: (Elucidation) A role *b* is a realizable entity such that *b* exists because there is some single bearer that is in some special physical, social, or institutional set of circumstances in which this bearer does not have to be & *b* is not such that, if it ceases to exist, then the physical make-up of the bearer is thereby changed

gUFO:ExtrinsicAspect: A **gufo:Aspect** that depends on one or more concrete individuals.

4.6.2 Examples and Considerations

BFO: Role: (Arp et al. 2015) the role of a portion of penicillin to serve as a drug, a lawyer, a nurse

SCO: indicator: A complex system component with an indicator role.

A role brings about some result or end through its manifestation, without being essential to its bearer. It is not essential *to Jim that he is a nurse, or to Mary that she is bankrupt* (Arp et al., 2015). Roles are externally grounded realizable entities, and do not change the physical make-up of their bearers if they cease to exist.

UFO’s hierarchy of Types accounts for Roles as *relationally dependent types, capturing relational properties shared by instances of a given kind*. Ideally, the Transformation Pattern for Roles would propose more complex modeling choices including the instantiation of Types, which is outside the scope of this work.

4.6.3 Transformation Patterns

Transformation Pattern #17 `BF0:Role` has target hierarchy `gUFO:ExtrinsicAspect`, respecting the following constraint:

- If the source ontology has concepts modeled as `BF0:Role` and as `BF0:Function` that are transformed to `gUFO:ExtrinsicAspect`, they must be *disjoint* subclasses of `gUFO:ExtrinsicAspect`

4.7 BF0:Function

4.7.1 Definitions

BF0:Function: (Elucidation) A function is a disposition that exists in virtue of its bearer’s physical make-up & this physical make-up is something the bearer possesses because it came into being either through evolution (in the case of natural biological entities) or through intentional design (in the case of artifacts) in order to realize processes of a certain sort

gUFO:ExtrinsicAspect: A `gufo:Aspect` that depends on one or more concrete individuals.

4.7.2 Examples and Considerations

BF0: Function: (Arp et al. 2015) the function of amylase in saliva to break down starch into sugar, the function of a pen to write

While Transforming a concept that is defined as `BF0:Function`, it is important to mind that UFO does not classify its dependent individuals according to their realization through processes. As a consequence, the Transformation Pattern suffers from loss of specificity in the transformed concepts.

4.7.3 Transformation Patterns

Transformation Pattern #18 `BF0:Function` has target hierarchy `gUFO:ExtrinsicAspect`, respecting the following constraint:

- If the source ontology has concepts modeled as `BFO:Role` and as `BFO:Function` that are transformed to `gUFO:ExtrinsicAspect`, they must be *disjoint* subclasses of `gUFO:ExtrinsicAspect`

4.8 BFO:GenericallyDependentContinuant

4.8.1 Definitions

BFO:GenericallyDependentContinuant: (Elucidation) A generically dependent continuant is an entity that exists in virtue of the fact that there is at least one of what may be multiple copies which is the content or the pattern that multiple copies would share

gUFO:Aspect: A `gufo:Endurant` that depends on at least one other concrete individual for its existence. A `gufo:Aspect` is a characteristic or trait of a concrete individual that is itself conceived as an individual.

4.8.2 Examples and Considerations

BFO: Generically Dependent Continuant: The pdf file on your laptop; the pdf file that is a copy thereof on my laptop; the sequence of this protein molecule; the sequence that is a copy thereof in that protein molecule; the content that is shared by a string of dots and dashes written on a page and the transmitted Morse code signal; the content of a sentence; an engineering blueprint

GeoCore: geological structure: a generically dependent continuant that is the pattern of a non-atomic geological object's internal arrangement.

SCO: indicator value: A representational information content entity that generically depends on some indicator and is about, providing insights into, the complex system to which the indicator belongs.

Generic dependence is not formalized in UFO's documentation. Existing literature proposes the idea of Modes and Social constructs that could provide the basis to model concepts such as *information artifacts*, represented in BFO by Generically-Dependent Continuants.

4.8.3 Transformation Patterns

Transformation Pattern #19 `BFO:GenericallyDependentContinuant` *has target hierarchy* `gUFO:Aspect` that is *disjoint* from the Aspects that represent the Specifically Dependent Continuants.

5 Occurrent Hierarchy

The final Transformation Patterns clarifies the Transformation Patterns from concepts defined as instances or as subclasses of the **BFO:Occurrent** classes, namely **BFO:Process**, **BFO:ProcessBoundary**, **BFO:SpatiotemporalRegion** and **BFO:TemporalRegion**. Figure 4 illustrates, on the left side, the excerpt of the BFO ontology covered by the Transformation Patterns defined in this document, along with the excerpt of OWL’s implementation of UFO ontology (on the right side) for which such Transformation Patterns will result in target hierarchies.

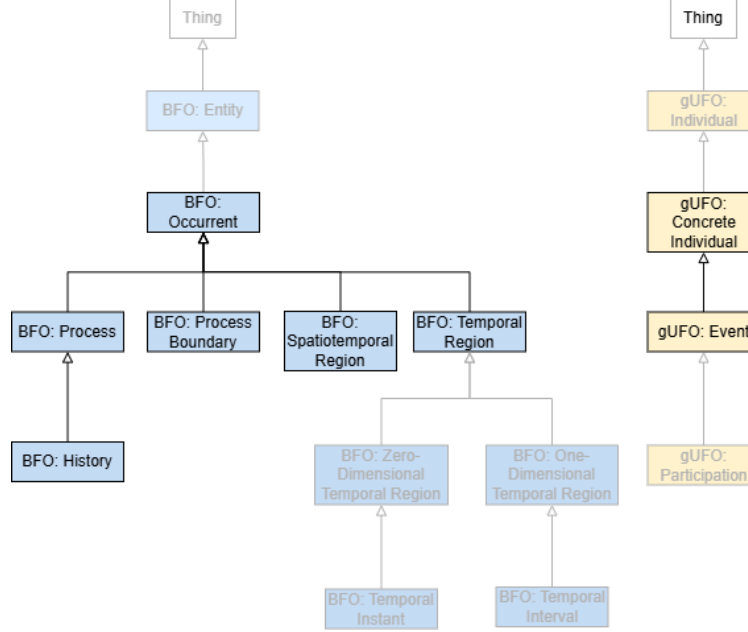


Figure 4: Excerpts of the taxonomies of Basic Formal Ontology (BFO) and Unified Foundational Ontology (UFO) covered by the Transformations Pattern #20

For each Transformation Pattern, we present the definition for BFO source and UFO target hierarchy of classes stated in the ontologies’ available documentation on GitHub. Additional examples (when available) and comments are provided to clarify the Transformation Pattern.

5.1 BFO:Occurrent

5.1.1 Definitions

BFO:Occurrent: (Elucidation) An occurrent is an entity that unfolds itself in time or it is the start or end of such an entity or it is a temporal or spatiotemporal region

BFO:Process: (Elucidation) p is a process means p is an occurrent that has some temporal proper part and for some time t , p has some material entity as participant

BFO:ProcessBoundary: p is a process boundary =Def p is a temporal part of a process & p has no proper temporal parts

BFO:SpatiotemporalRegion: (Elucidation) A spatiotemporal region is an occurrent that is an occurrent part of spacetime

BFO:TemporalRegion: (Elucidation) A temporal region is an occurrent over which processes can unfold

UFO:Event: A **gufo:ConcreteIndividual** that "occurs" or "happens" in time. They may be instantaneous or long-running. Events are those "things that happen to or are performed by" (Casati and Varzi, 2015) endurants.

5.1.2 Examples and Considerations

BFO: Process: (Arp et al. 2015) the life of this organism, that process of meiosis, the course of this disease, that flight of that bird, this fall of water down this waterfall

Geocore: geological process: It is a physical, or chemical, or biological, naturally occurring process that occurs on the Earth's surface or subsurface and occupies some geological time interval

SCO: interaction: A process that has at least two independent continuants (no spatial regions) as participants that have an effect upon one another.

SCO: system development: A process whose specific outcome is the progression of a system over time from an initial condition to a later condition.

BFO: Process Boundary: (Arp et al. 2015) the forming of a synapse, the onset of REM sleep, the final separation of two cells at the end of cell-division.

BFO: Spatiotemporal Region: (Arp et al. 2015) the region occupied by a human life, the region occupied by the development of a cancer tumor, the region occupied by a process of cellular meiosis

GeoCore: geological time interval: It is a BFO temporal interval that corresponds to a time interval within the Geological Time Scale.

Similarly to *Immaterial entities*, *Occurrents* in BFO include notions such as temporal boundaries, intervals and spatiotemporal regions that are not fully specified in UFO.

UFO's definition of Events imply the participation of Endurants, which would correlate to the more specific class of Processes in BFO. This motivates the Transformation Pattern proposed, that observes specific transformations of `BFO:Process`, `BFO:ProcessBoundary`, `BFO:TemporalRegion` and `BFO:SpatialRegion`.

5.1.3 Transformation Pattern

Transformation Pattern #20 `BFO:Occurrent` *has target Hierarchy* `gUFO:ConcreteIndividual`, considering the following constraints:

- If the source ontology has concepts modeled under the `BFO:Process` hierarchy, the `BFO:ProcessBoundary` hierarchy and the `BFO:TemporalRegion` hierarchy, they must be mapped to *disjoint* subclasses of `gUFO:Event`;
- `BFO:ProcessBoundary` *has target hierarchy* a `gUFO:Event` that *isEventProperPartOf* some other `gUFO:Event` from which it is disjoint.